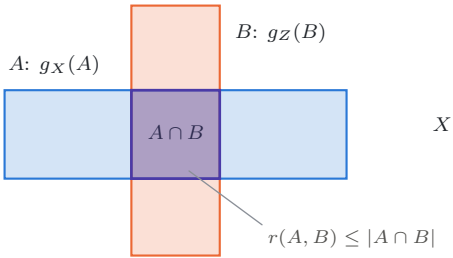
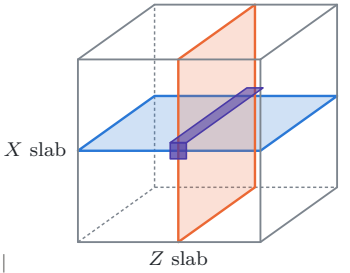
(a) Overlap-corrected cleaning



$$g_X(A) + g_Z(B) \le K + |A \cap B|$$

(b) Layer-code transverse slabs



$|A|,\ |B| = \Theta(d)$

$|A \cap B| = \Theta(K)$

(c) Structural consequence

slab supports: $\Theta(d)$ each

intersection: $\Theta(K)$

$$\chi_{XZ}(Q_N) = \Theta(K) = \Theta(n/d)$$

same slabs: fault-tolerant ports
+ structural no-go witness